11.4.3 Components of DBMS

The Database Management System can be divided into five major components as follows:

1.Hardware 2.Software

3.Data 4.Procedures/Methods

5.Database Access Languages

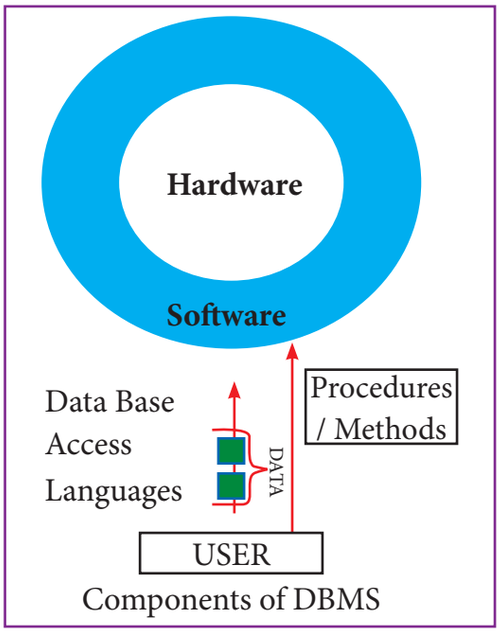

Figure 11.1

1. Hardware: The computer, hard disk, I/O channels for data, and any other physical component involved in storage of data

2. Software: This main component is a program that controls everything. The DBMS software is capable of understanding the Database Access Languages and interprets into database commands for execution.

3. Data: It is the resource for which DBMS is designed. DBMS creation is to store and utilize data.

4. Procedures/Methods: They are general instructions to use a database management system such as installation of DBMS, manage databases to take backups, report generation, etc.

5. DataBase Access Languages: They are the languages used to write commands to access, insert, update and delete data stored in any database.

Examples of popular DBMS: Dbase, FoxPro

11.5 Database Structure

Table is the entire collection of related data in one table, referred to as a File or Table where the data is organized as row and column.

Each row in a table represents a record, which is a set of data for each database entry.

Each table column represents a Field, which groups each piece or item of data among the records into specific categories or types of data. Eg. StuNo., StuName, StuAge, StuClass, StuSec.

A Table is known as a RELATION

A Row is known as a TUPLE

A column is known as an ATTRIBUTE

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